Hyperspectral Anomaly Detection With Adaptive Parameter Determination via Topological Disassembly Guided Pixel-Level Self-Information

Xiaotong Sun, Lina Zhuang, *Member, IEEE*, Lianru Gao, *Senior Member, IEEE*, Qingkang He, Andrea Marinoni, *Senior Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*—Hyperspectral anomaly detection presents persistent challenges due to the limitations of existing methodologies. Conventional methods based on signal processing typically employ fixed model assumptions and operate within rigid analytical modes, resulting in limited adaptability to diverse imaging scenes and constrained feature extraction capabilities. Meanwhile, machine learning-based methods generally necessitate annotated training samples, parameter optimization, and computationally intensive model training, which restricts their practical implementation in practical scenarios. To address these challenges, this article introduces an innovative perspective through the integration of point-set topology with information theory to analyze anomaly-background distribution characteristics within an imaging scene. This theoretical foundation enables enhanced characterization and extraction of discriminative information between these two distinct land covers. Adhering to a model-free design philosophy with data-adaptive parameter determination, we propose a novel method for hyperspectral anomaly detection, designated as parameterized method based on topological disassembly guided pixel-level self-information (PTDPS). The implementation of PTDPS comprises three fundamental stages: initially, a hyperspectral image (HSI) undergoes disassembly across spectral bands to generate parallel topological spaces; subsequently, the self-information associated with each pixel is evaluated using topological properties to quantify its uncertainty relative to the whole image, yielding initial anomaly scores; and ultimately, these initial scores are refined through a weighting strategy based on topological deviation from predominant data populations. The proposed PTDPS establishes a universal detection scheme applicable to diverse hyperspectral data. Experimental validation confirms that our method delivers outstanding detection accuracy while maintaining competitive computational efficiency, demonstrating robust adaptability and generalization performance across varied imaging scenes. To facilitate reproducibility and future research, the source code for this work has been made publicly available at: https://github.com/sxt1996/PTDPS**

***Index Terms*—Adaptive parameter determination, anomaly detection, hyperspectral image (HSI), model-free, point-set topology, self-information.**

I. Introduction

AS AN advanced remote sensing technology, hyperspectral imaging combines imaging and spectroscopy to generate a spectrum for each spatial pixel. Data acquisition across the visible, near-infrared, and shortwave infrared regions (0.4–2.5 μm) via contiguous narrow bands allows for the creation of a volumetric data cube, which forms the foundation for sophisticated quantitative analyses in various applications [1], [2]. The distinctiveness of the objects of interest depends on spatial distribution and spectral characteristics compared to the surrounding background [3], [4], [5]. This establishes operational conditions for target detection based on the investigation of spectral features. In hyperspectral remote sensing, target detection serves as a crucial technique for identifying objects of interest through differences in spectral responses across diverse materials within a specific scene [6], [7], [8]. Variations in physical and chemical properties among materials induce subtle differences in spectral features [9], [10]. Leveraging the abundant spectral-dimensional details embedded in hyperspectral data, hyperspectral target detection can precisely identify the target indistinguishable by conventional optical remote sensing technologies.

According to the availability of a priori information about the objects of interest, hyperspectral target detection primarily falls into two classes: supervised and unsupervised, namely, matching detection and anomaly detection. Given the frequent absence of well-informed spectral libraries in reality, anomaly detection, which can operate without any a priori information, has been widely used [11], [12], [13]. Hyperspectral anomaly detection provides robust technical support in fields such as geological survey, vegetation monitoring, damage assessment, and camouflage deception [14], [15], [16]. The development of related research holds valuable practical significance for both military and civilian sectors [17], [18], [19].

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Hyperspectral anomaly detection mainly comprises two methodological paradigms stemming from distinct disciplines: signal processing and machine learning [6], [9]. Since the 1990s, the signaling theory has been extensively applied to the processing and analysis of hyperspectral images (HSIs) [20], and the multitude of algorithms stemming from it has laid a solid foundation for the field [21], [22]. These algorithms primarily include those based on primitive spectral features, probability distribution statistics, and subspace projection. Algorithms based on primitive spectral features employ metrics such as spectral angles, distances, and similarity coefficients for matching [23], [24]. Algorithms based on probabilistic statistical models draw from binary hypothesis testing in statistics, assuming that testing pixels obey different multivariate normal distributions under different hypotheses. The technical difficulty of these algorithms lies in accurately estimating the parameters of models to achieve high detection accuracy. Representative algorithms include the Reed–Xiaoli detector (RXD) and the Bayesian Gaussian mixture model split-based anomaly detector (BGMMS) [25], [26]. Subspace projection-based algorithms enhance anomalies by constructing dedicated anomaly or background subspaces [27], thereby suppressing the background and maximizing the signal-to-noise ratio (SNR) for improved performance. Typical algorithms in this category include orthogonal subspace-projection (OSP) version of go decomposition (OSP-GoDec) [28], background joint sparse representation detection (BJSRD) [29], and low-rank and sparse matrix decomposition (LRaSMD) with OSP-based background suppression and adaptive weighting (LwOaW) [30].

By virtue of its core functionality in analyzing complex data and deciphering underlying patterns, machine learning can automatically construct and optimize models, effectively mine deep-level data features, and significantly enhance the quality of intelligent interpretation for HSIs. Hence, it has been instrumental in significantly accelerating progress in the field. To address challenges of the increasing complexity of data and higher requirements for applications, numerous novel and effective solutions have emerged from machine learning. Representation reconstruction, metric learning, kernel methods, tensor decomposition, and deep learning have become research hotspots, each demonstrating unique advantages and great application potential. Representation reconstruction-based methods leverage low-rank characteristics of hyperspectral data to reconstruct HSIs through techniques such as dictionary learning and sparse coding. The anomaly is detected by calculating reconstruction errors and anomalous pixels, typically corresponding to larger errors. Representative algorithms include the collaborative representation-based detector (CRD) [31], the deep self-representation learning framework (DSLF) [32], the sparse representation and linear mixture model (SR-LMM) [33], and LRaSMD [34]. By adaptively learning similarity metrics between samples and optimizing objective functions, metric learning-based methods can be effective in separating the anomaly and the background. These methods adapt well to complex data to improve detection accuracy. Notable algorithms include support vector data description (SVDD) [35], subspace selection-based discriminative forest (SSDF) [36], and tree topology-based anomaly detection (TTAD) [37]. By mapping data into high-dimensional feature spaces, kernel methods can capture nonlinear features in hyperspectral data. Techniques such as kernel principal components analysis (KPCA), kernel-independent component analysis (KICA), and SVDD are widely used in anomaly detection to enhance the capability of mining nonlinear data features [38], [39]. Representative algorithms include nonlinear spectral–spatial composite kernel-based detector (SSCKD) [40], kernel isolation forest-based hyperspectral anomaly detection (KIFD) [41], and SVDD detector [42]. In tensor decomposition-based methods, HSIs are represented as high-order tensors to deeply analyze the high-dimensional data structure and extract information for detection [43], [44]. Representative algorithms include low average rank with a total variation regularization model for hyperspectral anomaly detection (LARTVAD) [43], adaptive low-rank transformed tensor (ALRTT) [18], and structure tensor and guided filter (STGF) [45]. Deep learning methods have shown irreplaceable advantages in hyperspectral anomaly detection [46], [47], [48], [49]. Neural networks, including convolutional neural networks (CNNs) [50], autoencoders (AEs) [51], and generative adversarial networks (GANs) [52], excel at automatically learning hierarchical feature representations from raw data. This capability is particularly advantageous for analyzing complex hyperspectral data due to its high-dimensional and nonlinear spectral–spatial characteristics. Representative algorithms include CNN-based detector (CNND) [50], blind-spot self-supervised learning network (BS^3LNet) [53], and learning discriminative reconstruction based on constrained GAN (HADGAN) [52]. However, within the unsupervised anomaly detection paradigm, these data-driven approaches face several fundamental challenges. First, their high model complexity often conflicts with the scarcity of anomalies and the diversity of backgrounds in real HSIs, potentially leading to overfitting or poor generalization to novel application scenes. Second, the feature learning process, while powerful, can be a "black box" with learned representations not explicitly optimized for distinguishing the anomaly from the complex background, raising concerns about interpretability and reliability. Third, many models implicitly rely on assumptions regarding data distribution or anomaly characteristics, which may not hold universally. Consequently, their performance can be sensitive to application-specific parameter tuning. To address these core challenges, an ideal solution would benefit from a design that minimizes reliance on intricate, trainable models and explicit prior assumptions while determining its operational parameters in a principled and data-adaptive manner to ensure robustness and broad applicability. Accordingly, our work is motivated by this direction.

In summary, hyperspectral anomaly detection has been extensively studied in technical literature. Most existing methods exhibit robust theoretical foundations and high interpretability, as well as high efficiency. However, it is also true that all these methods rely on fixed models and assumptions (either in terms of metrics, representation, or data mapping techniques) for anomaly detection processing. On the one hand, this aspect results in low operational costs, hence enhancing the integration of these schemes in wider

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frameworks for remote sensing data analysis and application scenarios. This is particularly effective when investigating remote sensing scenes characterized by a slowly varying distribution of land covers. On the other hand, relying on fixed models and parameters reduces the ability of these algorithms to address the challenges of modern hyperspectral anomaly detection. This is caused by the high complexity of the scenes to be analyzed in operational scenarios, which leads to highly unbalanced datasets, high variability, and uncertainty associated with each data point, and limited sets of data points to be used for making the models learn and identify the anomaly.

To address the limitations imposed by fixed model assumptions and strict operational conditions, this article investigates the mathematical-statistical properties and distribution characteristics of the anomaly and the background. By leveraging the broad theoretical migration space provided by cross-disciplines such as hyperspectral remote sensing, point-set topology, and information theory, an adaptively parameterized method based on topological disassembly guided pixel-level self-information (PTDPS) is proposed. In particular, we adopt a model-free design philosophy with data-adaptive parameter determination to enhance adaptability to imaging scenes, reduce computational costs, and improve generalization capabilities in practical applications.

In particular, the principal novelty of this work lies in the following aspects.

Through the deep migration of point-set topology theory, the proposed method deconstructs anomaly-background distribution characteristics within an imaging scene from a novel perspective, thereby achieving a model-free design. By overcoming the limitations of traditional model-driven methods, including fixed modeling mechanisms and rigid analytical modes, we provide a flexible detection solution to adapt to real scenes with heterogeneous distributions of land covers.

Guided by topological disassembly, we construct parallel topological spaces and innovatively estimate self-information at the pixel level based on point-set properties. This facilitates an in-depth capture and extraction of the information differences embedded in the anomaly and the background land covers.

To address the limitations of a large number of existing methods with strict operational conditions, the proposed method achieves data-adaptive parameter determination. By eliminating the reliance on massive annotated samples and complex model-training processes, we significantly enhance the generalization capability of PTDPS. Consequently, even with limited computational resources, the proposed method maintains superior detection performance.

This article is systematically organized to guide the reader through the research process. It commences with a detailed elaboration of the proposed methodology in Section II. The discussion then transitions into Section III, which is dedicated to a comprehensive presentation and critical analysis of the experimental results. The study culminates with Section IV, where the principal findings are summarized, and concluding remarks are offered.

II. Proposed Topological Disassembly Guided Method Based on Pixel-Level Self-Information (PTDPS)

Fig. 1 illustrates the flowchart of PTDPS, and the methodology is elaborated in three main components: 1) the bandwise topological disassembly is implemented for an HSI under exam to construct parallel topological spaces; 2) the self-information of a random event associated with a testing pixel is estimated in each topological space, and an initial anomaly score is assigned to the testing pixel; and 3) the initial score is weighted based on the deviation degree of the testing pixel relative to a large and dense predominant population.

It is noteworthy that the theoretical foundations of point-set topology and information theory employed in this work have direct physical correspondences in hyperspectral anomaly detection. First, the calculated self-information of a pixel quantitatively measures its spectral rarity, which inversely corresponds to its occurrence probability within the overall scene. A high self-information value indicates a low occurrence probability, which is the defining characteristic of the anomaly that is spectrally distinct and spatially scarce. Second, the cardinality of a point set in the constructed topological space reflects the abundance of the corresponding material. A large cardinality signifies that many pixels share a high degree of spectral similarity, corresponding to a high abundance (and thus a high occurrence probability) of that land cover class, typically the background. Conversely, a small cardinality indicates spectral uniqueness and low material abundance, characteristics associated with a low occurrence probability and definitive of the anomaly. Thus, the core mechanism of PTDPS, which partitions pixels by spectral similarity into point sets and evaluates self-information based on cardinality, essentially operationalizes the detection of anomaly with low occurrence probability against the prevalent background. This provides a physically grounded and explainable detection framework.

A. Construction of Parallel Topological Spaces Guided by Topological Disassembly

As an abstract concept, topology can be utilized to describe geometric relationships [54], [55]. Point-set topology distinguishes itself from mathematical disciplines such as mathematical analysis, advanced algebra, and differential equations by emphasizing analytical methodologies over formula derivation and computational manipulation [56], [57]. This focus grants it enhanced applicability in practical problem-solving scenarios [58]. In this article, based on the point-set topology theory, we investigate mathematical-statistical properties of a research object by analyzing geometric relationships between point sets within the topological space.

An original HSI (i.e., dataset) can be structured as a 3-D data cube, where each pixel (i.e., data point) at each location corresponds to a spectrum. For pixels belonging to various classes of land covers, the feature differences in terms of spectral response are essentially reflected in numerical differences of various classes of spectra in specific bands.

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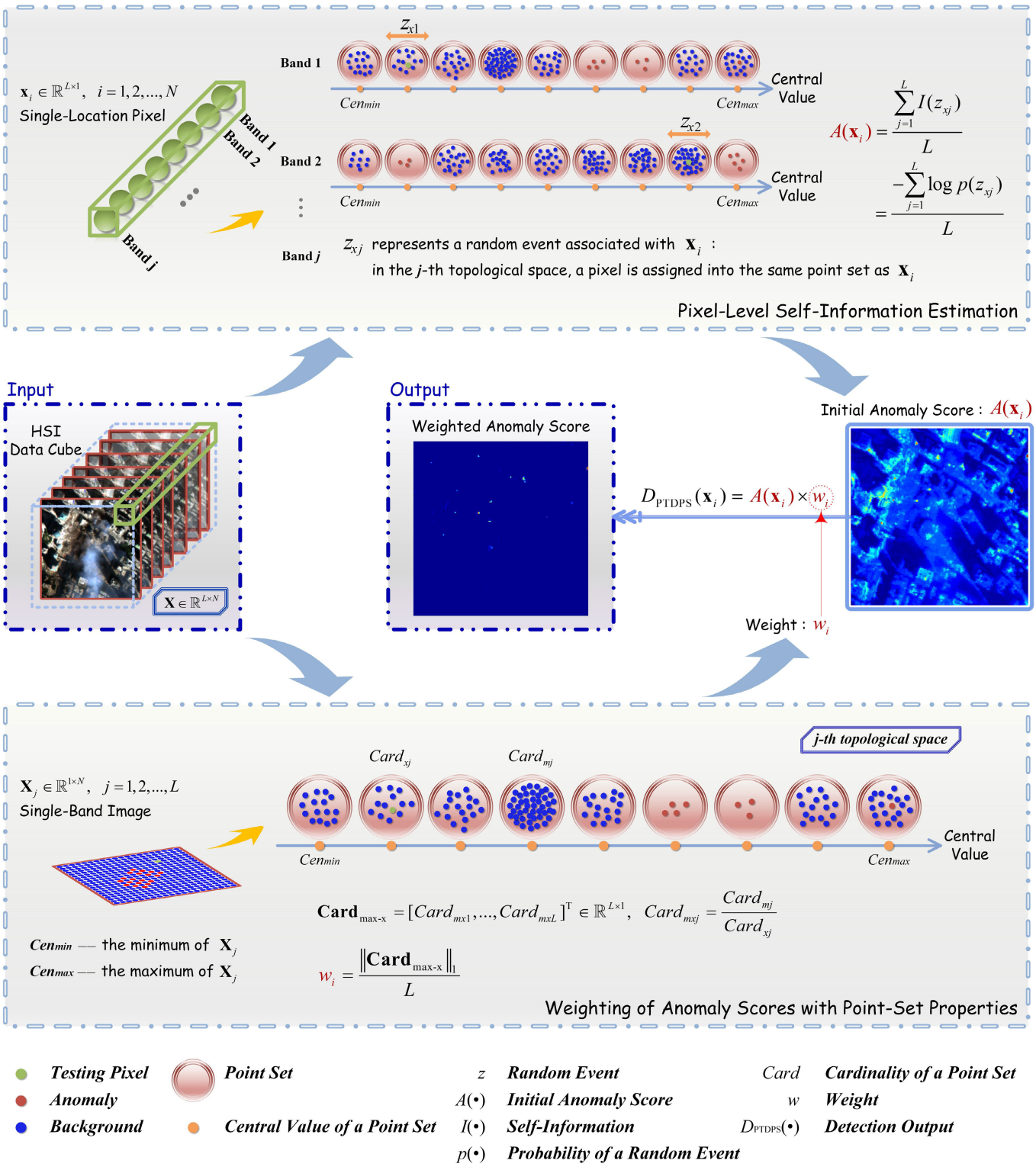


Fig. 1. Flowchart of the proposed PTDPS for hyperspectral anomaly detection.

In terms of numerical magnitude relationships, pixels in an original image exhibit a completely random and chaotic arrangement.

The core principle of applying point-set topology lies in disassembling these unstructured data points into multiple point sets to construct a topological space, where the data points can be rearranged. In the constructed topological space, data points with different mathematical-statistical properties are located in point sets with different properties [56], [57]. Consequently, data features that are latent and unquantifiable in the original data are mapped into a topological representation where they manifest as intuitive and quantifiable

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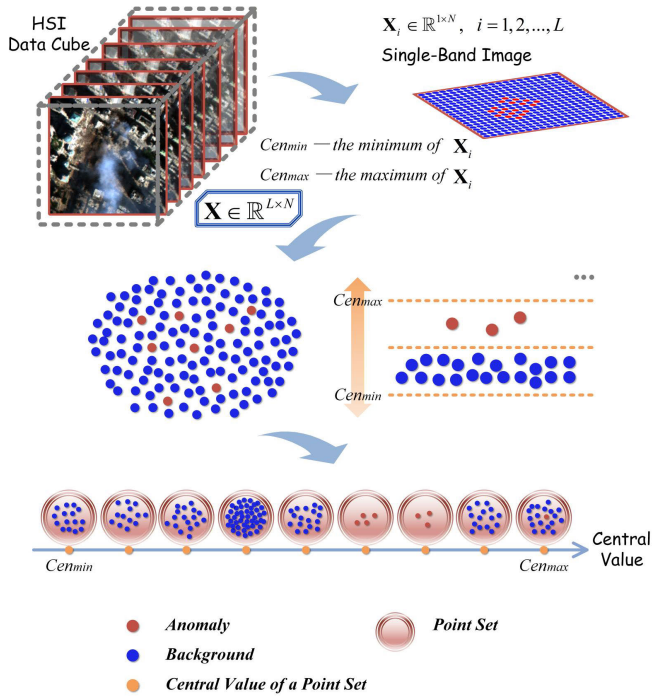


Fig. 2. Illustration of the construction of a topological space guided by topological disassembly.

point-set properties, enabling precise numerical comparison and objective interpretation.

This transformation reveals deep-level data features, thereby facilitating the development of algorithms tailored to specific application requirements. As a result, instead of relying on fixed model assumptions and rigid analytical modes, we can directly disassemble HSI data to construct the topological space and mine the data features for detection purposes.

In an imaging scene suitable for detection, the various land covers can be merely dichotomised into the anomaly and the background classes. The distribution of these two is extremely uneven. Specifically, an HSI can be treated as a collection of data points, where the background corresponds to data points that are densely distributed and larger in quantity, representing large and dense populations. The anomaly corresponds instead to data points that are sparsely distributed and smaller in quantity, representing small and sparse populations. Based on such differences in mathematical-statistical properties, we construct topological spaces in a specific form to achieve an anomaly-background separation by generating point sets with different properties.

Let $\mathbf{X} \in \mathbb{R}^{L\times N}$ denote an HSI under exam, where L refers to the number of spectral bands and N refers to the number of pixels. $\mathbf{X}$ can be interpreted as a set of L single-band images $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_L] \in \mathbb{R}^{L\times N}$, where $\mathbf{X}_i \in \mathbb{R}^{1\times N}$. It is known that feature differences in terms of spectral response between the anomaly and the background are embodied in specific bands. Thus, we perform the topological disassembly and construct parallel topological spaces band by band. The feature differences can be fully emphasized and mined.

For the construction of parallel topological spaces guided by topological disassembly, we operate with a single-band image $\mathbf{X}_i \in \mathbb{R}^{1\times N}, i = 1, 2, \ldots, L$, at a time. The interval between the minimum and maximum values of the spectral features in $\mathbf{X}_i$ is equally divided into multiple bins: each bin is identified by a central value. All pixels in $\mathbf{X}_i$ are assigned to the corresponding intervals based on spectral proximity to the central value of each bin. This process generates a collection of point sets, each corresponding to a central value and a set of pixels closest to it, as illustrated in Fig. 2. The central values corresponding to the anomalies and background are provided merely as examples and do not suggest any trends or patterns in the distribution ranges of their respective values. In the process of constructing a single topological space, it is possible for certain point sets to contain a mixture of pixels from both anomaly and background classes, as visualized in Fig. 2. This occurs because the initial partitioning is based solely on spectral proximity and topological connectivity, without any a priori information. While such mixed sets represent a minority, they could theoretically lead to misclassification if not addressed.

To mitigate the potential risk of misclassification stemming from the occasional mixing of pixels in individual topological sets within any single band, the final anomaly detection leverages the consistency of spectral responses across the entire data cube. Crucially, the pixelwise anomaly score is not derived from any single spectral band but is computed as the average of the self-information estimates obtained across all available bands. This design is grounded in the rationale that while an anomaly and a background pixel may coincidentally be grouped into the same point set in one band due to local spectral proximity, it is statistically less probable for this misclassification to persist consistently across numerous, distinct spectral bands where their inherent spectral similarity likely differs. Therefore, the multiband averaging operation serves as a robust error-compensation mechanism. It diminishes the contribution of random, band-specific misclassification errors by integrating evidence from the full spectral profile, ensuring that the anomaly score predominantly reflects the persistent abnormality of a pixel across the spectrum. This designed strategy effectively enhances the robustness and confidence of the detection results against the inherent local ambiguities present in unsupervised topological disassembly.

It should be emphasized that the number of point sets in each topological space is temporarily determined by a parameter e, which will be adaptively determined by data intrinsic properties. On this basis, topological disassembly is implemented by processing each band. Finally, the number of parallel topological spaces is equal to L. The detailed implementation steps for the topological disassembly guided construction of parallel topological spaces (TDPTSs) are outlined in Algorithm 1.

As described above, the construction of parallel topological spaces is accomplished through the direct disassembly at the data level, facilitating the implementation of model-free philosophy. As previously stated, the feature differences between various land covers are expressed through numerical differences. In each topological space, point sets are arranged following the rule of increasing central values, and the intervals between central values of any two neighboring point sets are equal. As a result, data points with small

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Algorithm 1 TDPTS (X, e)

Input: $\mathbf{X}=[\mathbf{x}_1,\ldots,\mathbf{x}_N]=[\mathbf{X}_1;\ldots;\mathbf{X}_L]\in\mathbb{R}^{L\times N}$, where $\mathbf{x}_i\in\mathbb{R}^{L\times 1}$, $\mathbf{X}_i\in\mathbb{R}^{1\times N}$ -the HSI under exam, e -the number of point sets in each topological space.

Output: $\mathbf{Card}\in\mathbb{R}^{L\times e}$ -the cardinalities of point sets in parallel topological spaces, $\mathbf{Card}_{\max}\in\mathbb{R}^{L\times 1}$ -the maxima of the cardinalities of point sets in parallel topological spaces.

Initialization: $\mathbf{Card}=[\mathbf{Card}_1;\ldots;\mathbf{Card}_L]\in\mathbb{R}^{L\times e}$, where $\mathbf{Card}_i=[Card_{i1},\ldots,Card_{ie}]\in\mathbb{R}^{1\times e}$, $\mathbf{Card}_{\max}=[Card_{m1},\ldots,Card_{mL}]^{\mathrm{T}}\in\mathbb{R}^{L\times 1}$.

1: Perform topological disassembly on the single-band image via a traversal to construct L parallel topological spaces;

for $i=1:L$ **do**

2: $\mathbf{X}_i\in\mathbb{R}^{1\times N}\leftarrow i$-th band of $\mathbf{X}$;

3: The minimum-maximum value range of $\mathbf{X}_i$ is partitioned into e equal intervals, each interval corresponding to a central value;

4: Assign all elements of $\mathbf{X}_i$ to the corresponding intervals according to their closest central values to generate e point sets;

5: Record cardinalities of e point sets in $\mathbf{Card}_i$;

6: Find the maximum of the cardinalities of e point sets in the current topological space and record it in $Card_{mi}$;

7: **end for (step 1)**

8: **return** $\mathbf{Card}$, $\mathbf{Card}_{\max}$

numerical differences are located in the same or adjacent point sets, whereas those with large numerical differences are located in distinct point sets with different properties that are clearly distinguishable. This imposes an order upon data points that were previously entangled in the disordered original space, thus introducing a structural regularity that makes them readily distinguishable within the topological representation. This ordered representation, in turn, establishes a rigorous foundation for the subsequent calculation of anomaly scores.

B. Estimation of Pixel-Level Self-Information

From the perspective of information theory, the significant differences in mathematical-statistical properties between the anomaly and the background lead to distinguishable distribution characteristics relative to the whole image. An HSI suitable for detection satisfies the basic condition of uneven distribution of land covers. The background is characterized by predominant populations, which are large and dense, whereas the anomaly is small and sparse populations. Consequently, the occurrence of a background pixel constitutes a high-probability event with lower uncertainty relative to the whole image, while an anomaly pixel manifests as a low-probability event exhibiting higher uncertainty. This uncertainty can be scientifically quantified through the self-information measurement. Specifically, the amount of self-information can be

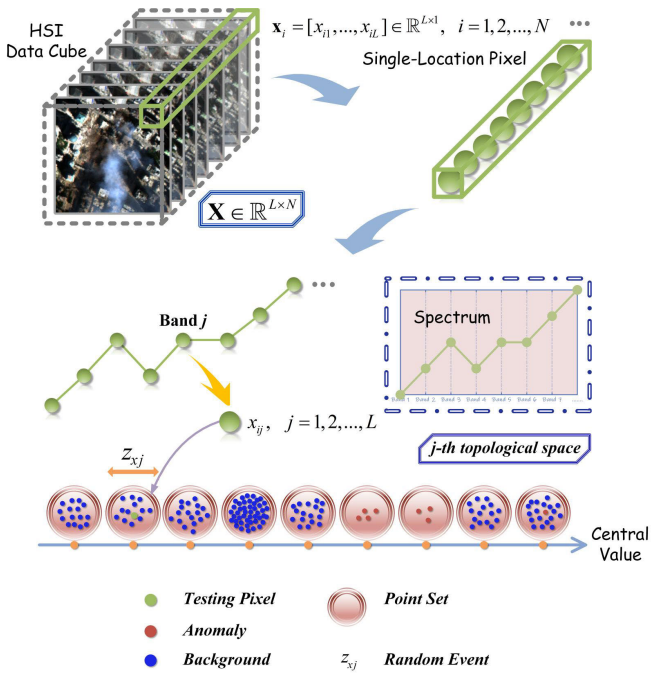


Fig. 3. Illustration of the generation of a random event for the estimation of pixel-level self-information.

estimated at the pixel level based on point-set properties in topological spaces. The results can be used to measure the uncertainty introduced by the occurrence of pixels belonging to different land covers relative to a whole image. Consequently, initial anomaly scores are assigned to testing pixels, enabling intuitive comparisons.

Through the operational mechanism of statistical inference, despite the deterministic construction of the aforementioned topological space, uncertainties inherent in its underlying stochastic process can nevertheless be reverse-inferred. In each topological space, all pixels of a single-band image are grouped into their respective point sets according to the central value to which their values are closest, thus generating random events. In the HSI $\mathbf{X}=[\mathbf{x}_1,\ldots,\mathbf{x}_N]\in\mathbb{R}^{L\times N}$, $\mathbf{x}_i\in\mathbb{R}^{L\times 1}$ is the ith testing pixel, and $\boldsymbol{Z}_i=\{z_{x1},\ldots,z_{xL}\}, i=1,2,\ldots,N$, denotes a set of random events. In $\boldsymbol{Z}_i$, message symbol $z_{xj}, j=1,2,\ldots,L$, represents a random event that may occur: in the jth topological space, a pixel is assigned to the same point set as $\mathbf{x}_i$. In Fig. 3, x_{ij} represents the positional value of $\mathbf{x}_i$ within the jth topological space, where it resides in a specific point set. The event of assigning a pixel to this same point set arises from a stochastic process. We treat this event as a random occurrence denoted by z_{xj}. A topological space is defined as a collection of point sets. The cardinality of a set refers to the number of elements it contains. Based on this, a set of probabilities $\boldsymbol{P}_i, i=1,2,\ldots,N$ of $\boldsymbol{Z}_i$ can be established

$$\begin{bmatrix}\boldsymbol{Z}_i\\ \boldsymbol{P}_i\end{bmatrix}=\begin{bmatrix}z_{x1} & z_{x2} & \cdots & z_{xL}\\ p(z_{x1}) & p(z_{x2}) & \cdots & p(z_{xL})\end{bmatrix} \tag{1}$$

where $p(z_{xj})$ denotes the probability of the event z_{xj} occurring, which can be calculated as follows:

$$p(z_{xj})=\frac{\mathrm{Card}_{xj}}{N} \tag{2}$$

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where Card_{xj}, $j = 1, 2, \ldots, L$, represents the cardinality of the point set where $\mathbf{x}_i$ is located in the jth topological space. Accordingly, the amount of self-information for each random event in $\boldsymbol{Z}_i$ can be estimated using the formula proposed by Shannon [59]

$$I\left(z_{xj}\right) = -\log p\left(z_{xj}\right) \tag{3}$$

where $I(z_{xj})$ is defined to measure the uncertainty associated with the occurrence of z_{xj}, with the logarithm base set to 2, and the unit of self-information is expressed in bits. By averaging the self-information of all random events in $\boldsymbol{Z}_i$, the initial anomaly score of $\mathbf{x}_i$ is obtained

$$A\left(\mathbf{x}_i\right) = \frac{\sum_{j=1}^{L} I\left(z_{xj}\right)}{L} = \frac{-\sum_{j=1}^{L} \log p\left(z_{xj}\right)}{L}. \tag{4}$$

As mentioned above, in all parallel topological spaces, the partitioning of pixels and $\mathbf{x}_i$ into the same point sets generates a set of random events $\boldsymbol{Z}_i$. The self-information of these events characterizes the uncertainty contained in the occurrence of $\mathbf{x}_i$. Therefore, the greater $A(\mathbf{x}_i)$ is, the higher the uncertainty of $\mathbf{x}_i$, which aligns more closely with the distribution characteristics of the anomaly in an imaging scene. Conversely, the smaller $A(\mathbf{x}_i)$ is, the lower the uncertainty of $\mathbf{x}_i$, which is more consistent with the distribution characteristics of the background. By traversing all pixels, initial anomaly scores are assigned to the entire image accordingly.

C. Weighting of Anomaly Scores With Point-Set Properties

The construction of a topological space on each band implements a rearrangement of the HSI data. In a specific band, pixels within the same point set exhibit minor numerical differences, indicating a higher similarity and a higher likelihood of belonging to the same class of land cover. The differences in mathematical-statistical properties between the anomaly and the background result in the partitioning of their respective pixels into point sets with different properties.

Specifically, within a topological space, the point sets containing the anomaly exhibit smaller cardinalities, indicating that fewer data points demonstrate high similarity to the anomaly. In contrast, the cardinalities of point sets corresponding to the background are relatively larger, indicating that a greater number of data points exhibit high similarity to the background. In the absence of any a priori information, pixels that differ from the majority in data features can be considered as exhibiting abnormality and have a higher probability of being judged as the anomaly. Therefore, the property of a point set, i.e., cardinality, can be utilized to measure the abnormality of a testing pixel.

In the HSI $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{L\times N}$, $\mathbf{x}_i = [x_{i1}, \ldots, x_{iL}] \in \mathbb{R}^{L\times 1}$ is the ith testing pixel, and we use $\mathbf{w} = [w_1, \ldots, w_N] \in \mathbb{R}^{1\times N}$ to record weights. According to the output of Algorithm 1, $\mathbf{Card} = [\mathbf{Card}_1, \ldots, \mathbf{Card}_L] \in \mathbb{R}^{L\times e}$ records the cardinalities of all point sets in L parallel topological spaces, where the cardinalities of e point sets in the jth topological space are recorded in $\mathbf{Card}_j = [\text{Card}_{j1}, \ldots, \text{Card}_{je}] \in \mathbb{R}^{1\times e}$. Accordingly, in the jth topological space, we obtain the cardinality of the point set in which x_{ij} is located and record it in Card_{xj}. Then, by traversing, the cardinalities of point sets where the testing pixel is located in all topological spaces are recorded in $\mathbf{Card}_x = [\text{Card}_{x1}, \ldots, \text{Card}_{xL}]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$. It is known that $\mathbf{Card}_{\max} = [\text{Card}_{m1}, \ldots, \text{Card}_{mL}]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$ records the maximum cardinalities for L topological spaces. The values of $\mathbf{Card}_{\max}$ and $\mathbf{Card}_x$ on each corresponding dimension are divided to obtain a set of ratios $\mathbf{Card}_{\text{max-x}} = [\text{Card}_{mx1}, \ldots, \text{Card}_{mxL}]^T \in \mathbb{R}^{L\times 1}$, where $\text{Card}_{mxj} = (\text{Card}_{mj}/\text{Card}_{xj})$. We use Card_{mxj} to quantify the deviation of $\mathbf{x}_i$ relative to the large and dense population within the jth topological space. Then, a weight is calculated

$$w_i = \frac{\|\mathbf{Card}_{\text{max-x}}\|_1}{L}. \tag{5}$$

The initial anomaly score is weighted to obtain the final detection output

$$D_{\text{PTDPS}}(\mathbf{x}_i) = A(\mathbf{x}_i) \times w_i. \tag{6}$$

For the testing pixel, the higher its uncertainty relative to the whole image and the greater its deviation from the large and dense populations, the more likely it is to be classified as an anomaly rather than background. Therefore, with a given threshold η, the following judgment criteria can be established:

$$D_{\text{PTDPS}}(\mathbf{x}_i) \begin{cases} > \eta \rightarrow & \text{anomaly} \\ < \eta \rightarrow & \text{background.} \end{cases} \tag{7}$$

The specific implementation steps outlined above are presented in Algorithm 2.

The integration of Algorithms 1 and 2 reveals a parameter e in the proposed method, which temporarily determines the number of point sets within a single topological space. Extensive experimental evidence indicates that assigning e a value strongly correlated with the HSI under exam enables implicit control over the parameter. This effectively eliminates explicit dependence on it, yielding the robust detection output. Specifically, we conduct a systematic investigation through experiments to quantitatively evaluate the impact of this parameter on detection performance. The experimental findings directly guide the evolution of our method toward a final, adaptively parameterized implementation.

III. Experimental Results and Discussion

For experiments, we select HSIs with diverse imaging scenes, varying sizes of the anomaly, and coverage of both natural and artificial objects as datasets. We adopt well-established qualitative and quantitative metrics to have the proposed method rigorously evaluated. Section III-A introduces HSI datasets, evaluation metrics, and experimental software-hardware configurations. Section III-B examines the impact of the aforementioned parameter, leading to its data-adaptive determination. This section also validates the effectiveness of the weighting strategy applied to anomaly scores. Comparative analyses are conducted in Section III-C to benchmark the proposed PTDPS against representative and cutting-edge methods, highlighting its effectiveness and superiority.

A. Experimental Preparation

1) Experimental Hyperspectral DataSets: Four publicly available benchmark hyperspectral datasets are selected. It is

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Algorithm 2 PTDPS (X, Card, $\mathbf{Card}_{\max}$)

Input: $\mathbf{X}=[\mathbf{x}_1,\ldots,\mathbf{x}_N]\in\mathbb{R}^{L\times N}$, where $\mathbf{x}_i\in\mathbb{R}^{L\times 1}$ -the HSI under exam, $\mathbf{Card}\in\mathbb{R}^{L\times e}$ -the cardinalities of point sets in parallel topological spaces, $\mathbf{Card}_{\max}\in\mathbb{R}^{L\times 1}$ -the maxima of the cardinalities of point sets in parallel topological spaces.

Output: $\mathbf{d}=[d_1,\ldots,d_N]\in\mathbb{R}^{1\times N}$ -detection results.

Initialization: $\mathbf{A}=[A_1,\ldots,A_N]\in\mathbb{R}^{1\times N}$, $\mathbf{Card}_x=[Card_{x1},\ldots,Card_{xL}]^{\mathrm{T}}\in\mathbb{R}^{L\times 1}$, $\mathbf{w}=[w_1,\ldots,w_N]\in\mathbb{R}^{1\times N}$.

1: Start a traversal to calculate the detection output;

for $i=1:N$ **do**

2: $\mathbf{x}_i\in\mathbb{R}^{L\times 1}\leftarrow i$-th testing pixel of $\mathbf{X}$;

3: Denote a set of random events by $\boldsymbol{Z}_i=\{z_{x1},\ldots,z_{xL}\}$, $i=1,2,\ldots,N$, where each event represents a pixel is assigned into the same point set as the testing pixel $\mathbf{x}_i$, and start accumulating the amount of self-information of all random events in $\boldsymbol{Z}_i$, $si=0$.

4: Start a traversal to estimate the self-information of the relevant random events in parallel topological spaces for the current testing pixel;

for $j=1:L$ **do**

5: Get the value of the testing pixel in the current topological space, $x_{ij}=\mathbf{x}_i(j)$;

6: Obtain the cardinalities of e point sets in the current topological space through **Card**, and find the point set where x_{ij} is located and record its cardinality in $Card_{xj}$;

7: With z_{xj} denoting the random event that a pixel is assigned into the same point set as the testing pixel $\mathbf{x}_i$, calculate the probability of its occurrence and its amount of self-information, $p(z_{xj})=\dfrac{Card_{xj}}{N}$, $I(z_{xj})=-\log p(z_{xj})$;

8: $si=si+I(z_{xj})$;

9: **end for (step 4)**

10: $A_i=\dfrac{si}{L}$;

11: Based on $\mathbf{Card}_{\max}$ and $\mathbf{Card}_x$, calculate a set of ratios to obtain $\mathbf{Card}_{\text{max-x}}\in\mathbb{R}^{L\times 1}$;

12: Calculate weight, $w_i=\dfrac{\|\mathbf{Card}_{\text{max-x}}\|_1}{L}$;

13: Weight the initial anomaly score obtained from the self-information estimation to obtain the detection output, $d_i=A_i\times w_i$;

14: **end for (step 1)**

15: **return d**

particularly noteworthy that these datasets contain no a priori information or labeled samples available, thereby ensuring a rigorous evaluation of the strengths of the proposed method under fully unsupervised conditions. The information about the experimental datasets, including illustrations, is meticulously presented in Table I.

The AVIRIS-WTC dataset was collected by NASA's Jet Propulsion Laboratory using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [60]. The dataset has a spatial dimension of 200×200 pixels, imaging the World Trade Center (WTC) area in New York. In addition, the dataset retains 224 spectral bands within the wavelength range of 370–2510 nm for research purposes.

The Nuance Cri hyperspectral sensor, characterized by a 10-nm spectral resolution, was used to collect the Cri dataset over a designated area at Wuhan University [61]. The entire image contains 400 × 400 pixels and 46 spectral channels.

Acquired by AVIRIS, the Segundo dataset covers the El Segundo area in California, USA, with a spatial resolution of 7.1 m and a size of 250 × 300 pixels. This configuration captures abundant spatial details [62]. In addition, this dataset includes 224 spectral channels spanning the wavelength range from 366 to 2496 nm. This particular scene contains diverse facilities such as a park, a petroleum refinery, residential areas, and a school district. Specifically, the distinct structures of the petroleum refinery, such as storage tanks and towers, are regarded as anomalies for subsequent processing and analysis.

The Nano dataset was acquired in 2021 using the Nano-Hyperspec sensor to image a mangrove area in TaiShan. It includes 273 spectral bands within the wavelength range of 400–1000 nm, with a spatial size of 220 × 220 pixels. In this scene, tarpaulins of different colors are regarded as the anomaly.

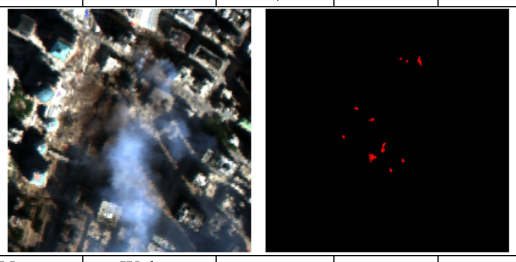
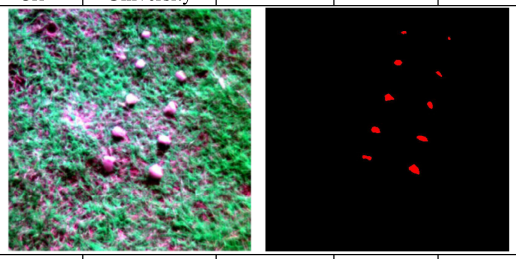
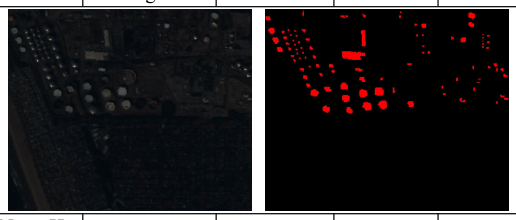
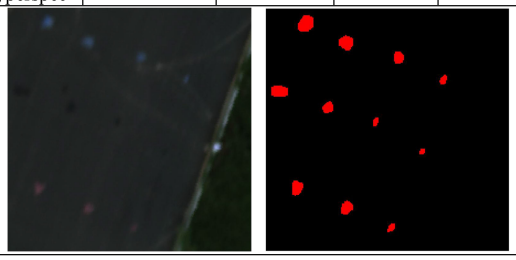
2) Evaluation Metrics: The evaluation metrics are widely recognized and applied in the field. These metrics enable in-depth analysis: qualitative analyses facilitate understanding the principles and mechanisms underlying the detection methods, while quantitative analyses enhance the objectivity and accuracy of the evaluation.

When evaluating the efficacy of a detection method, a set of specific thresholds is typically used to plot the receiver operating characteristic (ROC) curve [63]. As a graphical tool, the ROC curve clearly visualizes the relationship between probabilities of detections (PDs) and false alarm rates (FARs) under varying threshold η, as well as the correspondence between η and the FAR. Specifically, a method with superior performance should achieve a higher PD while maintaining a lower FAR. A pronounced shift of the $\mathrm{ROC}_{(\mathrm{FAR},\mathrm{PD})}$ curve toward the upper-left corner and the $\mathrm{ROC}_{(\eta,\mathrm{FAR})}$ curve toward the lower-left corner, thus, signifies enhanced detection performance.

The area under the curve (AUC) is a quantitative metric derived by aggregating the ROC curve into a single scalar value, which captures subtle performance differences between detection methods [64]. Specifically, a larger $\mathrm{AUC}_{(\mathrm{FAR},\mathrm{PD})}$ value and a smaller $\mathrm{AUC}_{(\eta,\mathrm{FAR})}$ value indicate better detection

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TABLE I
SPECIFICATIONS OF EXPERIMENTAL DATASETS

	Spectrometer	Capture Location	Spatial Resolution	Spatial Size	Number of Bands
AVIRIS-WTC	AVIRIS	New York	\	200×200	224
					
Cri	Nuance Cri	Wuhan University	\	400×400	46
					
Segundo	AVIRIS	El Segundo	7.1m	250×300	224
					
Nano	Nano-Hyperspec	TaiShan	\	220×220	273
					

AVIRIS: Airborne Visible/Infrared Imaging Spectrometer

performance of the corresponding method. Therefore, the AUC offers an intuitive standard for comparison.

The separability map, generated in conjunction with the ground-truth map, allows for a visual comparison of the value distributions corresponding to annotated anomaly and background regions [65]. The primary purpose of such a design is to provide a visual evaluation of the separation capability of a detection method. In the map, different colored boxes are used to represent the two distinct classes of land cover. In this representation, the anomaly class is marked by red boxes and the background by blue boxes. The less intersection between the two colored boxes corresponds to better separation capability of a detection method in complex imaging scenes.

3) Experimental Configurations: All experiments utilize identical software and hardware configurations. The MATLAB R2014a platform is adopted as the computational environment, offering extensive function libraries and toolboxes to accommodate diverse requirements from numerical computations to system-level simulations. All experiments are conducted on a standard workstation running the Windows 11 operating system to ensure computational reproducibility. The hardware configuration comprised a 12th-Generation Intel Core i7-1260 central processing unit (CPU), operating at a base frequency of 2.10 GHz, coupled with 16.0-GB GB of random access memory (RAM) to handle the computational workload. This configuration ensures computational efficiency and stability during intensive processing tasks, providing a reliable hardware infrastructure for experimental validation.

B. Adaptive Parameter Determination and Verification of Advantages

For the parameter e, a predefined range is tested across all datasets to expand experimental results and mitigate erroneous conclusions caused by randomness. The performance of the proposed PTDPS under varying e values is analyzed to assess its sensitivity and variation trends, thereby deriving a reliable and practical parameter determination scheme. The parameter e defines the number of point sets within a single topological space. A minimum of two point sets per topological space is required to ensure ordered data arrangements, while too many point sets could adversely impact the estimation of self-information with point-set properties in a topological space. The number of elements involved in generating random events for a testing pixel is known to be L, i.e., the number of bands. Hence, the setting range for e is from 2 to L, with an interval of 10, to observe whether the detection accuracy of the proposed PTDPS converges or not. Particularly, the experimental data are highly representative, featuring diversity in spatial size, number of bands, and land covers within imaging scenes. This contributes to establishing a parameter determination scheme of great reference value and thoroughly validating the robustness and superiority of the proposed PTDPS.

To assess the improvement gained by weighting the initial anomaly scores with the ratios of cardinalities within topological spaces, comparison experiments are conducted between detection outputs generated from weighted and initial anomaly scores. These results are evaluated through the quantitative metric AUC to thoroughly validate the enhancement in detection performance achieved by the weighting of anomaly scores with point-set properties.

1) Adaptive Parameter Determination: A quantitative evaluation of detection performance across parameter e within the predefined range is conducted. Fig. 4(a)–(d) presents line charts of AUCs for the proposed method on the AVIRIS-WTC, Cri, Segundo, and Nano datasets. Overall, except for minor local aberrations, the AUC gradually increases with e, accompanied by reduced magnitude of variation, eventually converging to stability. Observations from Fig. 4 indicate that PTDPS exhibits nearly insensitive performance across a substantially large parameter range, demonstrating that the explicit dependence on free parameter e can be eliminated. The parameter determination scheme is implemented by identifying the value of e at the data coordinates where the AUC curve begins to stabilize with minimal fluctuations. Across all

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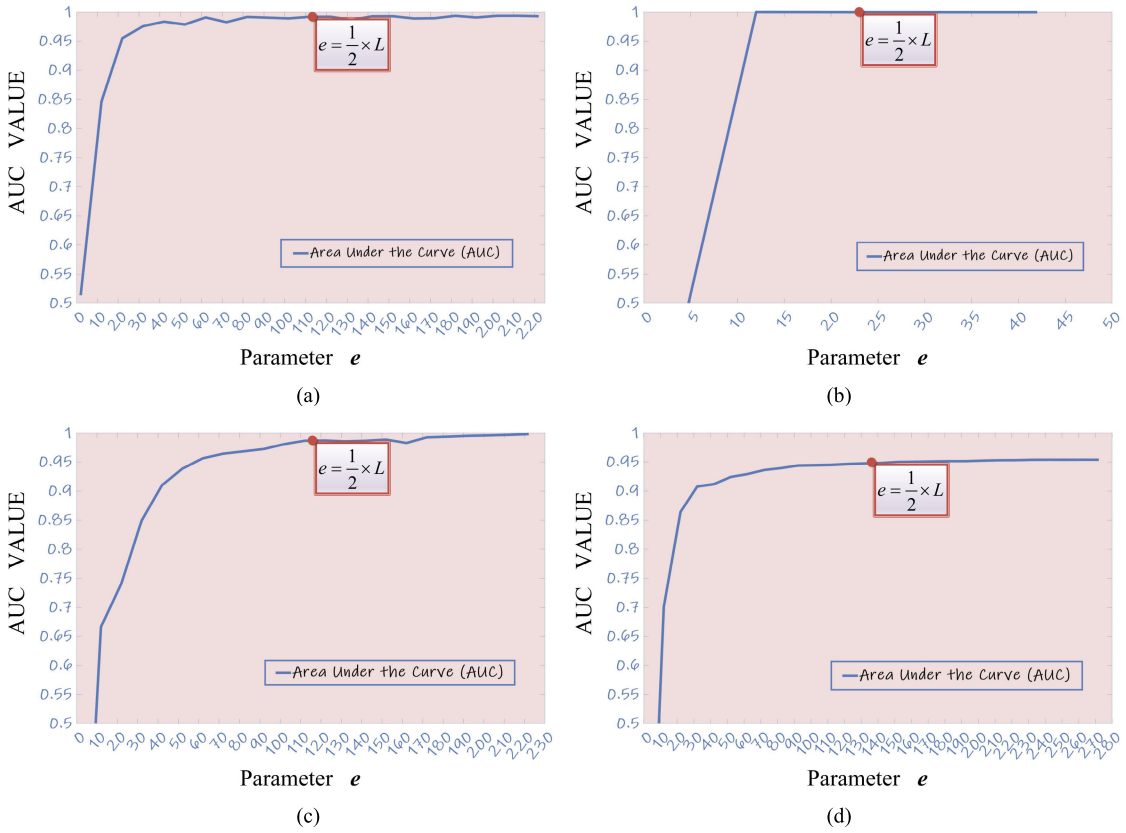


Fig. 4. Line chart depicting how the detection effect of PTDPS varies with different values of parameter e. (a) AVIRIS-WTC. (b) Cri. (c) Segundo. (d) Nano.

datasets, the AUC begins to converge at nearly (1/2) of the horizontal axis, leading to the setting of $e = (1/2) \times L$, while the value of L is inherently determined by the HSI under exam, meaning that we can leverage the intrinsic properties of the HSI itself to implicitly control the parameter. This approach reduces the uncertainty associated with manual parameter tuning and ultimately achieves the adaptive determination of the free parameter.

It is worth noting that each topological space contains e point sets, with each point set corresponding to a random event during statistical inference. The anomaly score calculation for each pixel involves L random events across all topological spaces. The configuration $e = (1/2) \times L$ achieves the robust detection output since, for an HSI under exam, when each topological space contains $(L/2)$ random events, the cumulative self-information of each pixel across L topological spaces suffices to quantify the abnormality of all pixels in the image. This finding aligns with the convergence phenomena observed across diverse datasets in Fig. 4(a)–(d). Consequently, PTDPS can be executed without independent parameter configuration in subsequent experiments, which conserves computational resources while significantly enhancing the practicality of the proposed method.

2) Verification for the Weighting of Anomaly Scores: As described in Section II-C, we quantify the abnormality of a testing pixel by measuring its deviation relative to large and dense populations, and correspondingly weight the initial anomaly scores to enhance the detection accuracy. The detection outputs corresponding to the initial and weighted anomaly scores across all experimental datasets are linearly stretched to the range 0–255 and visualized as detection maps by coloring pixels according to their magnitudes for visual comparison. Fig. 5 presents the comparison results on AVIRIS-WTC, Cri, Segundo, and Nano, respectively. In addition, the detection outputs, both weighted and unweighted, are evaluated separately. In Fig. 6, the obtained AUCs are plotted using bar charts with different colors and connected with star-marked lines for comparison. Based on the experimental evidence presented above, the principal conclusions are given as follows.

1) Fig. 5 visually demonstrates the enhanced effect of weighting, where the resulting detection maps show significantly improved prominence of anomaly compared to the unweighted outputs. Notably, the background regions demonstrate significantly lower values, demonstrating a stronger capability of suppressing the background, and the overall visual effect is better. In contrast, the detection maps for the unweighted anomaly scores suffer from severe false alarms, and in some cases, it is nearly impossible to discern the shapes and contours of the anomaly, resulting in a significantly degraded visual effect.
2) A clear performance gap is evident in Fig. 6, where the AUC of the weighted anomaly scores is substantially higher than the unweighted ones, thus verifying that the initial anomaly scores obtained from the self-information estimation fall short of high-precision detection requirements. The larger AUCs associated with the detection outputs of the weighted anomaly scores further demon-

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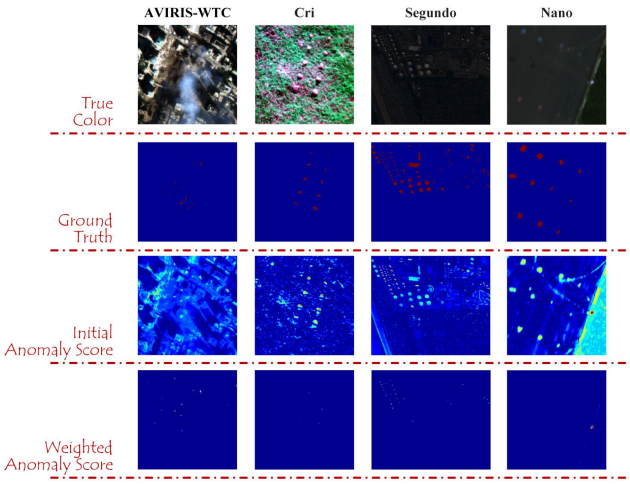


Fig. 5. Colored maps for comparison between detection outputs corresponding to initial and weighted anomaly scores.

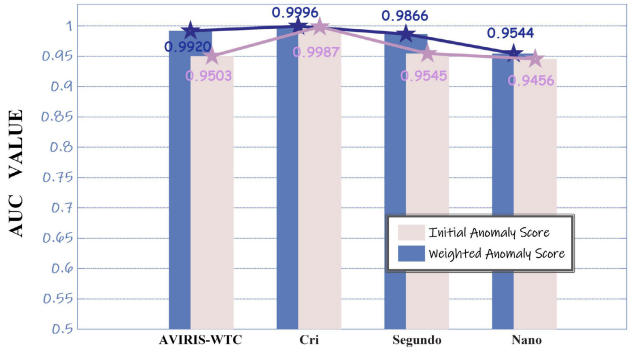


Fig. 6. Illustration of AUC evaluation results of detection outputs corresponding to initial and weighted anomaly scores.

strate that employing point-set properties for weighting effectively enhances the detection performance, endowing the proposed PTDPS with superior capabilities.

C. Comparison of Performance

In the comparison of detection performance, the most classic, advanced, and widely applied detection methods in the field are selected for benchmarking against the proposed PTDPS, thereby comprehensively validating its superior performance. The selected methods are listed as follows: 1) RXD (Global-RXD) [25]; 2) local RXD (Local-RXD) [63]; 3) robust principal component analysis RXD (RPCA-RXD) [66]; 4) CRD [31]; 5) low-rank and sparse representation-based detector (LRASR) [67]; 6) graph total variation regularized low rank representation-based detector (GTVLRR) [68]; 7) two-step GLRT (2S-GLRT) [69]; and 8) guided AE-based detector (GAED) [70]. In line with the earlier description, appropriate window sizes are configured per dataset to ensure an equitable comparison against methods using the concentric dual-window detection mode. On AVIRIS-WTC, the outer window size is set to 15×15, and the inner window size is set to 5×5. The window sizes for Cri are set to 25×25 and 15×15, respectively. The window sizes for Segundo are 21×21 and 11×11, respectively. The window sizes for Nano are 25×25 and 15×15. Throughout the experiments, all detection methods are implemented under strictly controlled conditions to secure robust and reproducible results, which lay the foundation for detailed and conclusive analyses.

1) Detection Effect: All detection results generated by the comparative methods undergo linear stretching to the range of 0–255. Fig. 7 presents the visualized detection maps. In summary, the findings support the following conclusions.

1) In terms of suppressing the background, Global-RXD, RPCA-RXD, LRASR, GTVLRR, and GAED demonstrate inadequate performance, as the contours of the background are distinctly visible in their detection maps. This issue is particularly pronounced on Nano, where the vegetation area in the lower right corner is clearly observable, resulting in a significant number of false alarms.
2) In terms of highlighting the anomaly, Local-RXD, CRD, and 2S-GLRT demonstrate poor performance. Compared to the ground-truth maps, these three methods exhibit varying degrees of disadvantage across all datasets except for Cri, making it challenging to visually identify the locations of anomalies.

Overall, PTDPS exhibits the best visual effect. Its detection result maps demonstrate effective background suppression, enabling clear identification of anomaly locations. This suggests that the proposed method achieves low FARs while also maintaining commendable PDs.

Fig. 8(a)–(d) illustrates the $\mathrm{ROC}_{(\mathrm{FAR,PD})}$ curves derived from the detection results of all methods applied to the AVIRIS-WTC, Cri, Segundo, and Nano datasets. The x-axis is displayed on a logarithmic scale to illustrate PDs at low FARs. To facilitate cross-comparison, Table II benchmarks the $\mathrm{AUC}_{(\mathrm{FAR,PD})}$ performance of all detection methods across the datasets, with the best results emphasized in bold. Furthermore, considering scenarios in practical applications that prioritize the FAR, Fig. 9(a)–(d) plots the $\mathrm{ROC}_{(\eta,\mathrm{FAR})}$ curves obtained by all methods on the experimental datasets, and Table III summarizes the corresponding $\mathrm{AUC}_{(\eta,\mathrm{FAR})}$ values. Qualitative and quantitative analyses of these evaluation results lead to the following conclusions.

1) In Fig. 8, GTVLRR, 2S-GLRT, and LRASR demonstrate the poorest performance on Cri, Segundo, and Nano, respectively, indicating that these three methods generally exhibit inferior detection effect. Although LRASR performs relatively well on Cri, this also underscores its instability in performance across various datasets.
2) The $\mathrm{ROC}_{(\mathrm{FAR,PD})}$ curves for Global-RXD, Local-RXD, RPCA-RXD, and CRD are intersecting and exhibit mediocre performance, showing neither proximity to the lower right corner nor to the upper left corner of the ROC space. Consequently, their performance differences are difficult to discern visually.
3) In contrast, the $\mathrm{ROC}_{(\mathrm{FAR,PD})}$ curves for GAED and PTDPS exhibit superior overall performance. Across all datasets, as the FAR increases, PTDPS achieves the highest PD, demonstrating a distinct advantage.
4) In the quantitative evaluation results presented in Table II, PTDPS achieves the highest $\mathrm{AUC}_{(\mathrm{FAR,PD})}$ on

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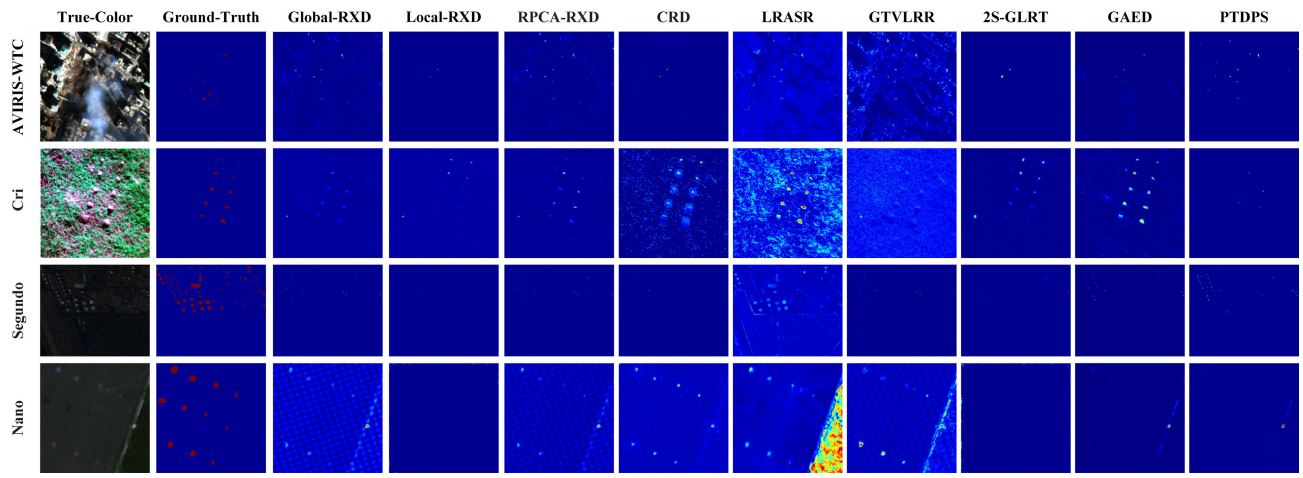


Fig. 7. Colored maps of detection methods under comparison.

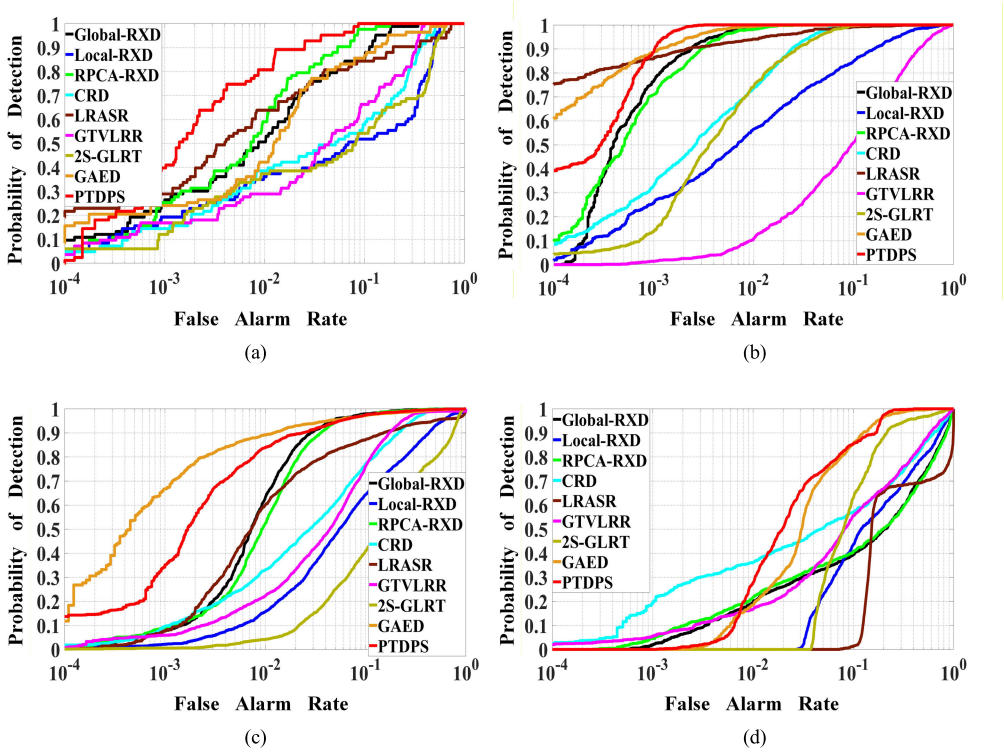


Fig. 8. $ROC_{(FAR,PD)}$ curves of detection methods under comparison. (a) AVIRIS-WTC. (b) Cri. (c) Segundo. (d) Nano.

TABLE II
EVALUATION RESULTS OF DETECTION METHODS UNDER COMPARISON

$AUC_{(FAR,PD)}$	Global-RXD	Local-RXD	RPCA-RXD	CRD	LRASR	GTVLRR	2S-GLRT	GAED	PTDPS
AVIRIS-WTC	0.9670	0.7180	0.9813	0.8677	0.9248	0.8939	0.7516	0.9521	**0.9920**
Cri	0.9990	0.9493	0.9987	0.9916	0.9956	0.8269	0.9900	0.9994	**0.9996**
Segundo	0.9841	0.8570	0.9810	0.9290	0.9287	0.9244	0.7791	**0.9874**	0.9866
Nano	0.6926	0.7390	0.6890	0.8039	0.6160	0.8013	0.8791	0.9478	**0.9544**
Average	0.9107	0.8158	0.9125	0.8981	0.8663	0.8616	0.8500	0.9717	**0.9832**

AVIRIS-WTC, Cri, and Nano, echoing the observations made in Fig. 8. The average $AUC_{(FAR,PD)}$ values listed in the final row indicate that PTDPS achieves the best overall performance across all datasets, and GAED performs second only to PTDPS, providing strong evidence that deep learning-based methods demonstrate strong competitiveness across experimental datasets with diverse imaging scenes.

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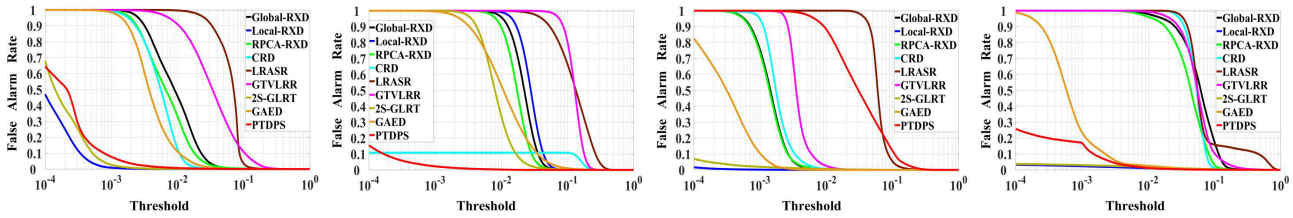


Fig. 9. $\mathrm{ROC}_{(\eta,\mathrm{FAR})}$ curves of detection methods under comparison. (a) AVIRIS-WTC. (b) Cri. (c) Segundo. (d) Nano.

TABLE III
EVALUATION RESULTS OF DETECTION METHODS UNDER COMPARISON

$\mathrm{AUC}_{(\eta,\mathrm{FAR})}$	Global-RXD	Local-RXD	RPCA-RXD	CRD	LRASR	GTVLRR	2S-GLRT	GAED	PTDPS
AVIRIS-WTC	0.0120	**0.0003**	0.0096	0.0064	0.0658	0.0481	0.0005	0.0055	0.0008
Cri	0.0237	0.0295	0.0189	0.0179	0.1488	0.1315	0.0103	0.0169	**0.0002**
Segundo	0.0017	**0.0001**	0.0017	0.0021	0.0641	0.0041	0.0003	0.0005	0.0480
Nano	0.0679	**0.0008**	0.0484	0.0553	0.1256	0.0654	0.0015	0.0021	0.0012
Average	0.0263	0.0077	0.0197	0.0204	0.1011	0.0623	**0.0032**	0.0063	0.0126

5) In Fig. 9, the $\mathrm{ROC}_{(\eta,\mathrm{FAR})}$ curves of Local-RXD are positioned closest to the lower-left corner on AVIRIS-WTC, Segundo, and Nano. This observation aligns with the results in Table III, where Local-RXD yields the smallest $\mathrm{AUC}_{(\eta,\mathrm{FAR})}$ values on these three datasets. Meanwhile, the $\mathrm{ROC}_{(\eta,\mathrm{FAR})}$ curves corresponding to 2S-GLRT exhibit outstanding performance across all datasets, and its average $\mathrm{AUC}_{(\eta,\mathrm{FAR})}$ value is the lowest among all methods. As for PTDPS, it achieves the best performance on Cri. Although its performance on Segundo is not particularly prominent, it remains highly competitive on all other datasets. Consequently, the final average of its $\mathrm{AUC}_{(\eta,\mathrm{FAR})}$ values ranks in the upper tier, demonstrating its strong capability in suppressing false alarms.

The left subfigures in Fig. 10(a)–(d) present a comparison of the statistical ranges for the detection results (anomaly versus background) achieved by different methods on the experimental datasets: AVIRIS-WTC, Cri, Segundo, and Nano. The subfigures on the right display these statistical ranges as box-and-whisker plots in two different colors. These observations lead to the following conclusions.

1) For the values at background locations, the detection results of Global-RXD, RPCA-RXD, LRASR, GTVLRR, and GAED exhibit a higher-than-average range, indicating inadequate background suppression capabilities, which may result in a substantial number of false alarms.
2) For the values at anomaly locations, the detection results of Local-RXD, CRD, and 2S-GLRT display a lower-than-average range, indicating poor capability in highlighting the anomaly, which can lead to suboptimal detection performance.
3) For the values at anomaly locations, the detection results of Local-RXD, CRD, and 2S-GLRT display a lower-than-average range, indicating poor capability in highlighting the anomaly, which can lead to suboptimal detection performance.
4) Overall, in the detection results of PTDPS, the values at background locations are suppressed to an extremely low level, while the values at anomaly locations are elevated to varying degrees, thereby enhancing the separability between the background and the anomaly. By employing a properly configured threshold, the proposed PTDPS can effectively stretch the anomaly away from the background, showcasing its remarkable efficacy in separating these two classes of land covers.

TABLE IV
TIME COMPLEXITY OF DETECTION METHODS UNDER COMPARISON

Method	Time Complexity	Definition of Variables
Global-RXD	$O(L^3+L^2N)$	L: the number of bands
Local-RXD	$O(L^3N+L^2N+Nwinout^2)$	N: the number of pixels
RPCA-RXD	$O(L^2N)$	
CRD	$O(L^3N)$	*winout*: the size of the outer window
LRASR	$O(LN)$	
GTVLRR	$O(N\log N)$	*win*: the size of the sliding window
2S-GLRT	$O(L^3N)$	
GAED	$O(LNwin^2+L^2NIter)$	*Iter*: the number of iterations
PTDPS	$O(L^2N)$	

2) Computational Efficiency: The comprehensive performance not only involves the detection effect but also the computational efficiency. Table IV provides the time complexity of all detection methods, while Table V summarizes the execution time from start to completion in seconds for each method. The last row presents the summed execution time across all datasets to facilitate an intuitive comparison. For clarity, all computational evaluations of the proposed method, including the asymptotic time complexity analyses and the recorded execution times, are based on its complete implementation. This encompasses the entire two-stage process, which comprises both Algorithms 1 and 2. This ensures a fair and holistic comparison of the computational efficiency against other comparative methods. Notably, the actual execution time depends on specific individual operations within each method. Thus, even methods with identical theoretical time complexity may exhibit significant runtime variations due to factors such as dynamic memory operations and noncontiguous data access patterns.

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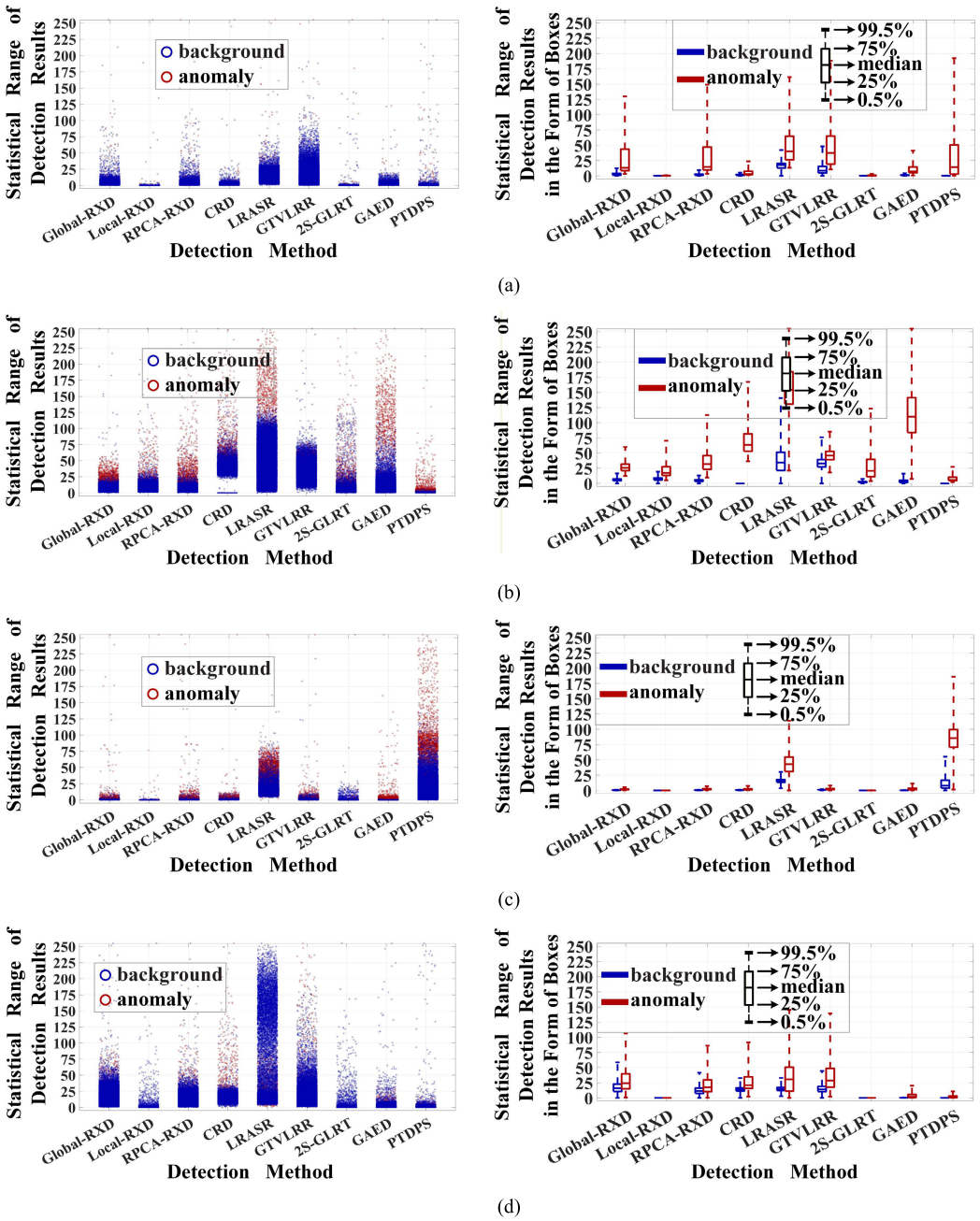


Fig. 10. Separability map of detection methods under comparison. (a) AVIRIS-WTC. (b) Cri. (c) Segundo. (d) Nano.

TABLE V
EXECUTION TIME OF DETECTION METHODS UNDER COMPARISON

	Global-RXD	Local-RXD	RPCA-RXD	CRD	LRASR	GTVLRR	2S-GLRT	GAED	PTDPS
AVIRIS-WTC	0.4072	323.7182	34.9560	665.2552	602.4300	456.3524	340.2546	263.1002	48.7944
Cri	0.4009	197.3978	18.8857	12496.8231	486.4512	2569.0856	16218.6591	722.5079	47.8323
Segundo	0.7001	752.2851	51.7122	1743.6284	137.5579	616.3960	1274.3321	367.9637	91.6919
Nano	0.7234	838.7336	57.2686	2491.7522	384.1284	468.6639	2266.6064	258.7577	71.8257
Sum	2.2316	2112.1347	162.8225	17397.4589	1610.5675	4110.4979	20099.8522	1612.3295	260.1443

Observations indicate that the computational efficiency of PTDPS ranks among the top few, being on par with traditional signal processing-based methods and significantly outperforming machine learning-based methods. Coupled with its previously validated superior detection effect, this demonstrates that PTDPS achieves the optimal performance

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in comparisons. Furthermore, the proposed method eliminates the requirements for parameter tuning, sample annotation, and model training, substantially saving manual effort in processing stages. Its excellent comprehensive performance endows it with great potential for widespread applications in practice.

IV. Conclusion

This work aligns with the trend toward high-performance computing by adopting a model-free design philosophy with data-adaptive parameter determination for hyperspectral anomaly detection. It overcomes limitations imposed by fixed model assumptions and strict operational conditions through the deep migration of point-set topology theory, establishing a flexible, robust, and universal solution for hyperspectral anomaly detection across diverse imaging scenes. Specifically, the construction of parallel topological spaces guided by topological disassembly facilitates the extraction of information differences between the anomaly and the background. On this basis, we innovatively estimate the pixel-level self-information using point-set properties to quantify the uncertainty of testing pixels relative to the whole image, thereby generating initial anomaly scores. To improve detection accuracy, the initial anomaly scores are weighted based on the deviation of testing pixels relative to large and dense populations within topological spaces. Experimental results demonstrate that the proposed PTDPS strikes an excellent balance between the detection effect and the computational efficiency, enabling a precise identification of the anomaly across diverse imaging scenes. Crucially, PTDPS operates effectively without relying on massive annotated samples and complex model-training processes, significantly reducing the consumption of computational resources while enhancing its generalization capability. This enables the proposed method to address various challenges associated with cross-scene and multiscale imaging conditions in practical applications while exhibiting superior detection performance.

It is important to acknowledge that this work represents a preliminary yet effective application of point-set topology, primarily utilizing cardinality as a key point-set property. Looking forward, significant potential lies in leveraging deeper topological invariants and advanced tools from modern topological analysis. For instance, concepts such as homology or persistent homology could be employed to analyze the global connectivity and structural features inherent in HSI data cubes. Such an analysis could reveal more intrinsic geometric and topological signatures of both the anomaly and the background, potentially leading to detectors with enhanced separability and interpretability. Subsequent work will also focus on integrating spatial information of the HSI data, for instance, by synergistically combining the proposed topological framework with a concentric dual-window detection mode. Such a hybrid strategy is anticipated to leverage both spectral separability and spatial coherence, thereby improving the stability and practical applicability of the proposed method across a wider range of scenarios.

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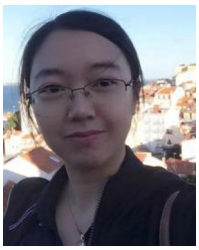
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